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United States Patent Application Publication

20250282329

Kind Code

A1

Publication Date

September 11, 2025

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FUEL CHARGING MODULE AND FUEL CHARGING METHOD

Abstract

A fuel charging module includes a storage tank that stores a fuel, a supply pipeline that supplies the fuel supplied from a charging station, to the storage tank, a flow rate control valve mounted on the supply pipeline and located on an upstream side of the supply pipeline with respect to a flow direction of the fuel, a heat-exchange pad contacting with the storage tank for exchange of heat with the storage tank, and a heat-exchange pipeline connected to the flow rate control valve, and that supplies the fuel to the heat-exchange pad. The flow rate control valve may be configured to determine the flow direction of the fuel supplied through the receptacle as a flow direction of the storage tank or the heat-exchange pad.

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Appl. No.: 18/949722

Filed: November 15, 2024

Foreign Application Priority Data

KR 10-2024-0031501

Mar. 05, 2024

Publication Classification

Int. Cl.: B60S5/02 (20060101); B67D7/32 (20100101); B67D7/36 (20100101)

U.S. Cl.:

CPC B60S5/02 (20130101); B67D7/32 (20130101); B67D7/36 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Korean Patent Application No. 10-2024-0031501, filed on Mar. 5, 2024, the entire contents of which is incorporated herein for all purposes by this reference.

BACKGROUND OF THE PRESENT DISCLOSURE

Field of the Present Disclosure

[0002] The present disclosure relates to a fuel charging module and a fuel charging method.

Description of Related art

[0003] Recently, as awareness of the crisis over the environment and depletion of oil resources has increased, research and development on fuel cell vehicles (hydrogen fuel cell vehicles), which are eco-friendly vehicles, have been highlighted.

[0004] A fuel cell vehicle is a vehicle using a fuel cell that utilizes hydrogen as a fuel. The fuel cell vehicle may be powered by electricity obtained by reacting stored hydrogen and oxygen. For the present reason, the fuel cell vehicle needs to be charged with hydrogen from a charging station.

[0005] The fuel cell vehicle may include a receptacle which is coupled to the charging station and a storage tank for storing hydrogen. Then, when the fuel cell vehicle is charged with hydrogen from the charging station, the hydrogen supplied through the receptacle may be accommodated in a compressed state into the storage tank.

[0006] Because hydrogen is accommodated in the storage tank while being compressed, an internal temperature of the storage tank may rise due to heat of compression of the hydrogen, and for safety reasons, charging of hydrogen is stopped when the temperature of the storage tank reaches a specific temperature.

[0007] Accordingly, the charging speed of hydrogen may decrease, and a demand for a fuel charging module and a fuel charging method which may optimize the charging speed of hydrogen while the temperature of the storage tank that does not reach a specific temperature is increasing.

[0008] The information included in this Background of the present disclosure is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.

BRIEF SUMMARY

[0009] Various aspects of the present disclosure are directed to providing a fuel charging module and a fuel charging method which may optimize a charging speed of a fuel while the safety of a storage tank is ensured.

[0010] The technical problems to be solved by the present disclosure are not limited to the aforementioned problems, and any other technical problems not mentioned herein will be clearly understood from the following description by those skilled in the art to which the present disclosure pertains.

[0011] According to an aspect of the present disclosure, a fuel charging module includes a storage tank that stores a fuel, a supply pipeline that supplies the fuel supplied from a charging station, to the storage tank, a flow rate control valve mounted on the supply pipeline, a heat-exchange pad contacting with the storage tank for exchange of heat with the storage tank, and a heat-exchange pipeline connected to the flow rate control valve, and that supplies the fuel to the heat-exchange pad, and the flow rate control valve is configured to supply the fuel supplied through the charging station, to the storage tank or the heat-exchange pad.

[0012] The heat-exchange pipeline may be inserted into an interior of the heat-exchange pad.

[0013] The storage tank may extend in a first direction, and a plurality of storage tanks may be

configured to be disposed in a second direction intersecting the first direction, and the heat-exchange pad may include accommodation recesses disposed on opposite sides of the plurality of storage tanks in a third direction intersecting the first direction and the second direction to accommodate the plurality of storage tanks, and support portions formed on sides of the accommodation recesses in the second direction to support sides of the storage tanks in the second direction, and into which the heat-exchange pipeline is inserted.

[0014] A plurality of support portions may be spaced apart from each other in the second direction while the accommodation recesses being interposed therebetween, and the heat-exchange pipeline may extend to pass through the plurality of support portions in the first direction.

[0015] The fuel charging module may further include a temperature detecting sensor which is configured to detect a temperature of the storage tank, and a processor that is configured to control the flow rate control valve so that a flow rate of the fuel supplied from the charging station to the storage tank is adjusted in response that the temperature of the storage tank reaches a reference temperature.

[0016] The processor is configured for controlling the flow rate control valve so that supply of the fuel to the storage tank is stopped and the fuel is supplied to the heat-exchange pipeline in response that the temperature of the storage tank reaches the reference temperature.

[0017] The processor is configured for controlling the flow rate control valve to increase the flow rate of the fuel supplied to the storage tank, and decrease the flow rate of the fuel supplied to the heat-exchange pipeline, in response that the temperature of the storage tank decreases to less than the reference temperature after the flow rate control valve is operated to supply the fuel to the heat-exchange pipeline.

[0018] The fuel charging module may further include a signal transmitter that transmits a signal to the charging station, and the processor may be configured to, control the signal transmitter to transmit a charging stop signal to the charging station, in response that the temperature of the storage tank is increased or maintained for a predetermined time period after the flow rate control valve is operated to supply the fuel to the heat-exchange pipeline.

[0019] The heat-exchange pipeline may include one end portion connected to the flow rate control valve, and an opposite end portion extending from the one end portion and formed on an opposite side to the one end portion, and the fuel charging module may further include a recovery pipeline connecting the opposite end portion of the heat-exchange pipeline and the charging station.

[0020] The heat-exchange pipeline may include one end portion connected to the flow rate control valve and an opposite end portion extending from the one end portion and formed on an opposite side to the one end portion, and the fuel charging module may further include a recovery pipeline connected to the opposite end portion of the heat-exchange pipeline, and a heat-exchange portion that exchanges heat between the fuel recovered through the recovery pipeline and the fuel supplied from the charging station.

[0021] The fuel charging module may further include a sub tank connected to the recovery pipeline, and which is spaced apart from the heat-exchange pad to accommodate the fuel recovered through the recovery pipeline, a sub pipeline connecting the sub tank and the storage tank to supply the fuel accommodated in the sub tank to the storage tank, and an opening/closing valve disposed on the sub pipeline.

[0022] The heat-exchange portion may be positioned on a location, where the supply pipeline crosses the sub pipeline.

[0023] According to another aspect of the present disclosure, a fuel charging method includes a connecting operation of connecting a receptacle to a charging station, a first fuel supplying operation of supplying a fuel from the charging station to a storage tank, a temperature measuring operation of measuring a temperature of the storage tank, and a tank cooling operation of controlling flow of the fuel supplied through the receptacle so that the fuel is directed to a heat-exchange pad in response that the temperature of the storage tank reaches a reference temperature

or is more than the reference temperature.

[0024] The tank cooling operation may include controlling the flow of the fuel so that all the fuel supplied through the receptacle is directed to the heat-exchange pad.

[0025] The fuel charging method may further include a second fuel supplying operation of controlling the flow of the fuel supplied to the storage tank to increase the flow rate of the fuel in response that the temperature of the storage tank decreases to less than the reference temperature.

[0026] The second fuel supplying operation may include increasing a flow rate of the fuel supplied to the storage tank, and decreasing a flow rate of the fuel directed to the heat-exchange pad.

[0027] The fuel charging method may further include a flow rate maintaining operation of controlling the flow of the fuel to maintain the flow rate of the fuel supplied to the storage tank and the flow rate of the fuel directed to the heat-exchange pad in response that the temperature of the storage tank is maintained for a predetermined time period after the second fuel supplying operation.

[0028] The fuel charging method may further include a signal transmitting operation of transmitting a charging stop signal to the charging station in response that the temperature of the storage tank increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature.

[0029] The fuel charging method may further include a notifying operation of providing a notification to a user in response that the temperature of the storage tank increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature.

[0030] The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 is a perspective view of a fuel charging module according to an exemplary embodiment of the present disclosure;

[0032] FIG. 2 is a schematic view exemplarily illustrating a state, in which a fuel is supplied to a storage tank according to an exemplary embodiment of the present disclosure;

[0033] FIG. 3 is a longitudinal sectional view exemplarily illustrating a state, in which a fuel is supplied to a heat-exchange pipeline according to an exemplary embodiment of the present disclosure;

[0034] FIG. 4 is a schematic view of a fuel charging module according to another exemplary embodiment of the present disclosure;

[0035] FIG. 5 is a schematic view of a fuel charging module according to another exemplary embodiment of the present disclosure;

[0036] FIG. 6 is a flowchart illustrating a fuel charging method according to an exemplary embodiment of the present disclosure; and

[0037] FIG. 7 is a flowchart illustrating a fuel charging method according to another exemplary embodiment of the present disclosure.

[0038] It may be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as included herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined in part by the particularly intended application and use environment.

[0039] In the figures, reference numbers refer to the same or equivalent portions of the present disclosure throughout the several figures of the drawing.

DETAILED DESCRIPTION

[0040] Reference will now be made in detail to various embodiments of the present disclosure(s), examples of which are illustrated in the accompanying drawings and described below. While the present disclosure(s) will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that the present description is not intended to limit the present disclosure(s) to those exemplary embodiments of the present disclosure. On the other hand, the present disclosure(s) is/are intended to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims.

[0041] Hereinafter, various exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In adding reference numerals to the components of the drawings, it is noted that the same components are denoted by the same reference numerals even when they are drawn in different drawings. Furthermore, in describing the exemplary embodiments of the present disclosure, when it is determined that a detailed description of related known configurations and functions may hinder understanding of the exemplary embodiments of the present disclosure, a detailed description thereof will be omitted.

[0042] Furthermore, in describing the components of the exemplary embodiments of the present disclosure, terms, such as first, second, “A”, “B”, (a), and (b) may be used. The terms are simply for distinguishing the components, and the essence, the sequence, and the order of the corresponding components are not limited by the terms. Unless defined differently, all the terms including technical or scientific terms include the same meanings as those generally understood by an ordinary person in the art, to which the present disclosure pertains. The terms, such as the terms defined in dictionaries, which are generally used, should be construed to coincide with the context meanings of the related technologies, and are not construed as ideal or excessively formal meanings unless explicitly defined in the present disclosure.

[0043] Hereinafter, various exemplary embodiments of the present disclosure will be described in detail with reference to FIGS. 1 to 7.

[0044] FIG. 1 is a perspective view of a fuel charging module according to an exemplary embodiment of the present disclosure. FIG. 2 is a schematic view exemplarily illustrating a state, in which a fuel is supplied to a storage tank according to an exemplary embodiment of the present disclosure; FIG. 3 is a longitudinal sectional view exemplarily illustrating a state, in which the fuel is supplied to a heat-exchange pipeline according to an exemplary embodiment of the present disclosure.

[0045] Referring to FIG. 1, FIG. 2, and FIG. 3, a fuel charging module **100** may be understood as a module which is mounted in an interior of a fuel cell vehicle (a hydrogen fuel cell vehicle) to receive hydrogen (hereinafter “a fuel”) which is the fuel of the fuel cell vehicle (hereinafter “a vehicle”) from a charging station **10** and store it.

[0046] The fuel charging module **100** may include a storage tank **110** which is configured to store the fuel, a receptacle **120** for connection to an external charging station **10**, a signal transmitter **130** which is formed adjacent to the receptacle **120**, and a supply pipeline **140** which is configured to supply the fuel supplied from the receptacle **120** to the storage tank **110**.

[0047] The storage tank **110** may extend in a first direction (the “X” direction or an opposite direction to the “X” direction), and a plurality of storage tanks **110** may be provided to be disposed in a second direction (the “Y” direction or an opposite direction to the “Y” direction) that intersects the first direction. The plurality of storage tanks **110** may be spaced apart from each other in the second direction thereof.

[0048] The receptacle **120** may be a component for direct connection to the charging station **10** for

the fuel charging module **100** to receive the fuel from the charging station **10**.

[0049] A signal transmitter **130** may be formed adjacent to the receptacle **120**. When the receptacle **120** and the charging station **10** are connected to each other, the signal transmitter **130** may transmit a connection signal of the receptacle **120** and the charging station **10** to the charging station **10**.

[0050] For example, The signal transmitter **130** may transmit IR (Infrared Rays). Receiver may be attached to the nozzle of the charging station. When the nozzle and the receptacle **120** are connected, information may be exchanged through the signal transmitter **130** and the nozzle. In this way, because the signal transmitter **130** and the nozzle must be within a range where signals can be communicated with each other, the signal transmitter **130** needs to be located around the receptacle **120**.

[0051] As another example, when the nozzle is fastened to the receptacle **120**, a network between objects can be connected through interfaces such as QR codes and barcodes around the receptacle **120**.

[0052] The signal transmitter **130** may transmit a ready signal of the fuel charging module **100** to the charging station **10** when the fuel charging module **100** is ready to receive the fuel from the charging station **10**.

[0053] Furthermore, the signal transmitter **130** may transmit a charging stop signal of the fuel charging module **100** to the charging station **10**, in a situation, in which the fuel charging module **100** may no longer receive the fuel from the charging station **10**. As an exemplary embodiment of the present disclosure, when a pressure of any one of the plurality of storage tanks **110** is more than a predetermined reference pressure, the signal transmitter **130** may transmit the charging stop signal to the charging station **10**. Accordingly, the predetermined reference pressure may be a pressure value excluding a specific error range from 350 bar*1.5.

[0054] The supply pipeline **140** may extend from the receptacle **120** to the storage tank **110**, and may be configured to supply the fuel supplied from the receptacle **120**, to the storage tank **110**.

[0055] The fuel charging module **100** may include a flow rate control valve **150** which is disposed on the supply pipeline **140** and is located on an upstream side of the storage tank **110** in a flow direction of the fuel.

[0056] The supply pipeline **140** may include a first supply pipeline **160** that connects the receptacle **120** and the flow rate control valve **150**, and a second supply pipeline **170** that connects the flow rate control valve **150** and the storage tank **110**.

[0057] A pressure sensor **165** configured to detect the pressure of fuel that flows through the first supply pipeline **160** may be provided on the first supply pipeline **160**.

[0058] The second supply pipeline **170** includes a main pipeline **171** which is connected to the flow rate control valve **150** and extends in the second direction, and a branch pipeline **172** which is branched from the main pipeline **171** and extends in the first direction.

[0059] The main pipeline **171** may extend between one end portion which is connected to the flow rate control valve **150**, and an opposite end portion which is connected to the branch pipeline **172** which is most distant from the flow rate control valve **150**.

[0060] A branch support portion **180** may support the second supply pipeline **170** at a portion, at which the main pipeline **171** and the branch pipeline **172** are connected to each other. The branch pipeline **172** may connect the main pipeline **171**, and an inlet **111** of the storage tank **110**.

[0061] A solenoid valve and the like may be formed in the inlet **111** of the storage tank **110**. The solenoid valve formed in the inlet **111** may be configured to determine a flow rate of the fuel supplied into each of the storage tanks **110**, or whether the fuel is supplied to an interior of each of the storage tanks **110**. Furthermore, a temperature detecting sensor configured to detect the temperature of the storage tank **110** may be provided in the inlet **111**.

[0062] Due to the present structure, the fuel charging module **100** may cause the fuel supplied through the receptacle **120** to pass through the first supply pipeline **160**, the second supply pipeline **170**, and the inlet **111** and flow into the storage tank **110**.

[0063] Meanwhile, the fuel that flows into an interior of the storage tank **110** may be stored in a compressed state to increase an amount of the stored fuel. To achieve this, a pressure may be applied to the fuel, and through the present process, compression heat which is heat generated as the fuel is compressed may be dissipated from the interior of the storage tank **110**. As the compression heat is radiated from the interior of the storage tank **110**, an internal temperature of the storage tank **110** may rise, and for the safety of the storage tank **110**, the storage tank **110** may be used only within a predetermined temperature range. As an exemplary embodiment of the present disclosure, the storage tank **110** may be used only within a range of -40 degrees Celsius to 85 degrees Celsius.

[0064] According to the fuel charging module **100** according to the exemplary embodiment of the present disclosure, the internal temperature of the storage tank **110** may be adjusted to improve a charging speed of the fuel charging module **100** while the storage tank **110** is used at a temperature within the above-described range.

[0065] The fuel charging module **100** may include a heat-exchange pad **200** that contacts with the storage tank **110** for exchange of heat with a plurality of storage tanks **110**, and a heat-exchange pipeline **300** which is connected to the flow rate control valve **150** and is configured to supply the fuel to an interior of the heat-exchange pad **200**.

[0066] The heat-exchange pad **200** may contact with the plurality of storage tanks **110** in a third direction (the “Z” direction or an opposite direction to the “Z” direction) that intersects the first and second directions of the plurality of storage tanks **110**. The heat-exchange pad **200** may be formed to receive the heat from the plurality of storage tanks **110** to lower the temperature of the plurality of storage tanks **110**.

[0067] The heat-exchange pad **200** may include a first heat-exchange pad **210** that exchanges heat with a plurality of storage tanks **110** on one side (the “Z” direction) of the plurality of storage tanks **110** in the third direction, and a second heat-exchange pad **220** that exchanges heat with the plurality of storage tanks **110** on an opposite side (an opposite direction to the “Z” direction) of the plurality of storage tanks **110** in the third direction thereof.

[0068] The first heat-exchange pad **210** and the second heat-exchange pad **220** may be formed to surround the plurality of storage tanks **110** on opposite sides of the plurality of storage tanks **110** in the third direction, respectively.

[0069] In more detail, the first heat-exchange pad **210** may include a first accommodation recess **211** which is disposed on one side of the plurality of storage tanks **110** in the third direction, and into which each of the storage tanks **110** is inserted, and a first support portion **212** which is formed on one side of the first accommodation recess **211** in the second direction thereof.

[0070] The second heat-exchange pad **220** may include a second accommodation recess **221** which is disposed on an opposite side of the plurality of storage tanks **110** in the third direction, and into which each of the storage tanks **110** is inserted, and a second support portion **222** which is formed on one side of the second accommodation recess **221** in the second direction thereof.

[0071] The first support portion **212** may define the first accommodation recess **211** on opposite sides of the first accommodation recess **211** in the second direction, and the second support portion **222** may define the second accommodation recess **211** on opposite sides of the second accommodation recess **221** in the second direction thereof.

[0072] A plurality of first support portions **212** and a plurality of second support portions **222** may be provided to be spaced apart from each other in the second direction while the first accommodation recess **211** and the second accommodation recess **221** being interposed therebetween.

[0073] The first accommodation recess **211** and the second accommodation recess **221** may be formed at positions corresponding to each other when viewed from the third direction thereof. Furthermore, the first support portion **212** and the second support portion **222** may be formed at positions corresponding to each other when viewed from the third direction thereof.

[0074] The plurality of storage tanks **110** may be inserted into and accommodated in the first accommodation recess **211** and the second accommodation recess **221**, respectively. The first support portion **212** and the second support portion **222** may be disposed on opposite sides of the plurality of storage tanks **110** in the second direction to support the opposite sides of the plurality of storage tanks **110** in the second direction thereof.

[0075] The heat-exchange pipeline **300** may be inserted into the first heat-exchange pad **210** and the second heat-exchange pad **220**. The heat-exchange pipeline **300** may include a first heat-exchange pipeline **310** which is inserted from the flow rate control valve **150** into the first heat-exchange pad **210**, and a second heat-exchange pipeline **320** which is inserted from the flow rate control valve **150** into the second heat-exchange pad **220**.

[0076] The first heat-exchange pipeline **310** may be inserted into an interior of the first heat-exchange pad **210** and extend to pass through each of the first support portions **212** of the first heat-exchange pad **210** in the first direction. The first heat-exchange pipeline **310** may first pass through, among the plurality of first support portions **212**, the first support portion **212** which is closest to the flow rate control valve **150**, and may extend to sequentially pass through, among the plurality of first supports **212**, the remaining first support portions **212**.

[0077] Similarly, the second heat-exchange pipeline **320** may be inserted into an interior of the second heat-exchange pad **220** and extend to pass through each of the second support portions **222** of the second heat-exchange pad **220** in the first direction. The second heat-exchange pipeline **320** may first pass through, among the plurality of second support portions **222**, the second support portion **222** which is closest to the flow rate control valve **150**, and may extend to sequentially pass through, among the plurality of second support portions **222**, the remaining second support portions **222**.

[0078] That is, the first heat-exchange pipeline **310** and the second heat-exchange pipeline **320** may extend to pass through the plurality of first support portions **212** and the plurality of second support portions **222** in the first direction. This is for increasing an area, by which the fuel that flows through the first heat-exchange pipeline **310** and the second heat-exchange pipeline **320** receives heat from the first heat-exchange pad **210** and the second heat-exchange pad **220**, respectively.

[0079] Meanwhile, the flow rate control valve **150** may supply the fuel supplied through the charging station **10** to the storage tank **110** or the heat-exchange pad **200**. In other words, the flow rate control valve **150** may be configured to determine a flow direction of the fuel to supply the fuel supplied through the receptacle **120** to the storage tank **110** and the heat-exchange pad **200**.

[0080] The fuel charging module **100** may include a controller **400** that is configured to control the flow rate control valve **150** including an actuator operatively connected to the controller **400**. The controller **400** may include memory and a processor. The memory may include a volatile memory, such as a static random access memory (S-RAM) and a dynamic random access memory (D-RAM) for temporarily storing data while electric power is supplied, and a nonvolatile memory, such as a read only memory (ROM) and an erasable programmable read only memory (EPROM) for preserving data even when supply of electric power is cut off.

[0081] The processor may include various logic circuits and operation circuits, may be configured for processing data according to a program provided from the memory and generate control signals according to the processing results.

[0082] When the receptacle **120** is completely connected to the charging station **10**, as illustrated in FIG. 2, the fuel may flow to the second supply pipeline **170** through the flow rate control valve **150**. Thereafter, the fuel may flow into the storage tanks **110**.

[0083] Thereafter, as an amount of the fuel which is accommodated in the interior of the storage tank **110** increases over time, the temperature of the storage tank **110** may increase. The temperature of the storage tank **110** may be detected by a temperature sensor at specific time intervals. When the temperature of the storage tank **110** is not a reference temperature or higher than the reference temperature even over time, a state of the flow rate control valve **150** may be

maintained. Here, a time difference may be about 10 ms.

[0084] The processor may detect the temperature of the storage tank **110** at intervals and may be configured to conclude that the temperature of the storage tank **110** is increased or decreased when the difference between the two temperatures deviates a predetermined error range.

[0085] Thereafter, when the temperature of the storage tank **110** detected by the temperature detecting sensor reaches a predetermined reference temperature, the processor is configured for controlling the flow rate control valve **150** to adjust the flow rate of the fuel supplied from the charging station **10** to the storage tank **110**.

[0086] The flow rate control valve **150** may include a sensor, a control unit, a driving unit and a mechanical regulator. The driving unit may be an actuator. The driving unit may be provided within the valve or may be mounted outside of the flow rate control valve **150**.

[0087] Sensor may monitor flow status of the fluid in real time by measuring flow rate, pressure, temperature, etc. Based on the data collected from the sensor, the control unit may determine whether to perform flow rate control using the PID control method. Afterwards, based on commands from the control unit, the driving unit may be operated electronically, such as an electric motor or solenoid. When the driving unit is activated, the mechanical regulator may adjust the degree of opening and closing of the flow rate control valve **150**.

[0088] In more detail, when the temperature of the storage tank **110** reaches the reference temperature, as illustrated in FIG. **3**, the processor is configured for controlling the flow rate control valve **150** so that the fuel supplied through the charging station **10** is directed to the first and second heat-exchange pipelines **310** and **320**. Accordingly, the reference temperature may be a temperature excluded from 85 degrees Celsius by a specific error range.

[0089] When the temperature of the storage tank **110** reaches the reference temperature or increases to the reference temperature or higher than the reference temperature, the processor is configured for controlling the flow rate control valve **150** to stop the supply of the fuel to the storage tank **110** and supply the fuel to the first and second heat-exchange pipelines **310** and **320**.

[0090] Accordingly, the flow rate control valve **150** may internally adjust opening amounts of openings that are fluidically communicating with the first and second heat-exchange pipelines **310** and **320** to 100%, and may adjust opening amounts of openings that are fluidically communicating with the second supply pipeline to 0%.

[0091] Due to the flow rate control valve **150**, the fuel may no longer be supplied to the plurality of storage tanks **110**, and heat may be exchanged between the plurality of storage tanks **110** and the first and second heat-exchange pads **210** and **220**. Thereafter, the heat may be transferred to the fuel that flows through the first and second heat-exchange pipelines **310** and **320**, through the first and second heat-exchange pads **210** and **220**. According to the present principle, the plurality of storage tanks **110** may transfer the heat to the fuel, and the temperature of the storage tank **110** may decrease.

[0092] When the temperature of the storage tank **110** decreases after the flow rate control valve **150** supplies the fuel to the first and second heat-exchange pipelines **310** and **320**, the processor is configured for controlling the flow rate control valve **150** to gradually increase the flow rate of the fuel supplied to the storage tank **110** again and gradually decrease the flow rate of the fuel supplied to the heat-exchange pipeline **300** again.

[0093] Accordingly, the flow rate control valve **150** may adjust the flow rate of the fuel supplied to the second supply pipeline **170** and the first and second heat-exchange pipelines **310** and **320** by adjusting opening amounts of the openings that are fluidically communicating with the second supply pipeline **170** and the first and second heat-exchange pipelines **310** and **320** internally, respectively.

[0094] Thereafter, the processor is configured for controlling the flow rate control valve **150** to maintain the flow rate of the fuel supplied to the storage tank **110** and the flow rate of the fuel supplied to the heat-exchange pipeline **300** when the temperature of the storage tank **110** is

maintained. In other words, the flow rate control valve **150** may internally maintain the opening amounts of the openings that are fluidically communicating with the second supply pipeline **170** and the first and second heat-exchange pipelines **310** and **320**, respectively.

[0095] When the temperature of the storage tank **110** increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature for a predetermined time period after the flow rate control valve **150** supplies the fuel to the first and second heat-exchange pipelines **310** and **320**, the processor is configured for controlling the signal transmitter **130** to transmit a charging stop signal to the charging station **10**. Here, the specific time period may be about 10 s.

[0096] By the present principle, the fuel charging module **100** can continuously receive the fuel until the fuel is supplied from the charging station **10** and the temperature of the storage tank **110** reaches the reference temperature. Thereafter, when the temperature of the storage tank **110** reaches the reference temperature or is more than the reference temperature, the fuel charging module **100** does not stop supplying the fuel, but rather may cause the fuel to flow to the first and second heat-exchange pipelines **310** and **320**. When the temperature of the storage tank **110** decreases to less than the reference temperature, the fuel charging module **100** may gradually increase the amount of the fuel supplied to the storage tank **110**, increasing an overall fuel charging speed.

[0097] When the temperature of the storage tank **110** does not decrease from the reference temperature, charging by the fuel charging module **100** may be stopped by transmitting a charging stop signal to the charging station **10** in the signal transmitter **130**.

[0098] Meanwhile, the fuel charging module **100** may further include a recovery pipeline **500** for recovering the fuel that flows through the first and second heat-exchange pipelines **310** and **320** to the charging station **10**.

[0099] Each of the first and second heat-exchange pipelines **310** and **320** may include one end portion which is connected to the flow rate control valve **150** and an opposite end portion extending from the one end portion and is formed on an opposite side to the one end portion. Accordingly, the recovery pipeline **500** may connect an opposite end portion of each of the first and second heat-exchange pipelines **310** and **320** and the charging station **10**.

[0100] FIG. **4** is a schematic view of a fuel charging module according to another exemplary embodiment of the present disclosure.

[0101] Referring to FIG. **4**, the fuel charging module **100-1** may further include a recovery pipeline **500** connected to opposite end portions of the first and second heat-exchange pipelines **310** and **320**, a sub pipeline **510**, an opening/closing valve **520**, a sub tank **600**, and a heat-exchange portion **700**, in addition to the components of the fuel charging module **100** illustrated in FIG. **1**. The description of FIG. **1** is used for the remaining components of the fuel charging module **100-1** of FIG. **4**.

[0102] The fuel charging module **100-1** does not recover the fuel from the heat-exchange pipelines **310** and **320** to cause it to flow to the charging station **10** as illustrated in FIG. **1**, but rather may recover the fuel from the first and second heat-exchange pipelines **310** and **320** and may supply it to the plurality of storage tanks **110** again.

[0103] The recovery pipeline **500** may extend from the first and second heat-exchange pipelines **310** and **320** to the sub tank **600**. The sub tank **600** may accommodate the fuel which is recovered through a sub inlet **601** connected to the recovery pipeline **500**. The sub tank **600** may be spaced apart from the heat-exchange pad **200** (see FIG. **1**).

[0104] The sub tank **600** may be connected to the sub pipeline **510**. The fuel accommodated in the interior of the sub tank **600** may flow through the sub pipeline **510**. The sub pipeline **510** may connect the sub tank **600** and the plurality of storage tanks **110** to supply the fuel accommodated in the sub tank **600** to the storage tanks **110**.

[0105] The sub pipeline **510** may connect the sub tank **600** and the plurality of storage tanks **110**, and a heat-exchange portion **700** and an opening/closing valve **520** may be provided on the sub

pipeline **510**.

[0106] In an exemplary embodiment of the present disclosure, the heat-exchange portion **700** may be positioned on a location, where the supply pipeline **160** crosses the sub pipeline **510**.

[0107] Accordingly, the heat-exchange portion **700** may be configured to exchange heat between the fuel recovered through the recovery pipeline **500** and the fuel supplied from the charging station **10** (see FIG. **1**). In more detail, in FIG. **4**, the heat-exchange portion **700** may be configured to exchange heat between the fuel that flows through the sub pipeline **510** and the fuel which is supplied from the charging station **10**. By the heat-exchange portion **700**, the temperature of the fuel that flows through the sub pipeline **510** may be lowered, and the temperature of the fuel which is supplied from the charging station **10** may be increased.

[0108] The opening/closing valve **520** may be provided with a solenoid valve, and the like, and may open or close an internal passage of the sub pipeline **510**. The opening/closing valve **520** may be controlled by a processor of the controller **400** (see FIG. **1**).

[0109] The heat-exchange portion **700** may be located on an upstream side of the opening/closing valve **520** with respect to a flow direction of the fuel that flows through the sub pipeline **510**. However, the present disclosure is not limited thereto, and the heat-exchange portion **700** may be located on a downstream side of the opening/closing valve **520**.

[0110] The fuel charging module **100-1** may store the fuel in the sub tank **600** separately from the plurality of storage tanks **110**, and then may supply the fuel from the sub tank **600** to the storage tank **110** when the fuel in at least one of the plurality of storage tanks **110** is insufficient or an internal pressure of at least one storage tank **110** is lower than the internal pressure of the other storage tanks.

[0111] By the present principle, a larger amount of fuel may be accommodated than in a structure with only the storage tank **110** in an interior of the vehicle, and a travel distance of the vehicle may be increased by preventing a pressure imbalance between the storage tanks **110**.

[0112] Meanwhile, the temperature of the fuel accommodated in the sub tank **600** may be higher than the temperature of the fuel which is introduced into the interior of the storage tank **110**, and thus, the fuel may be supplied to the storage tank **110** through the heat-exchange portion **700** at a relatively lower temperature.

[0113] The processor is configured to determine the amount of the fuel accommodated in each of the storage tanks **110** through the temperature of each of the storage tanks **110**. The temperature sensor and a tank pressure valve may be provided at each of the storage tank **110**. The temperature sensor may measure the temperature of the storage tank **110**. The pressure measured by each of the storage tanks **110** may be transmitted to the processor provided at the charging station **10**. The processor may calculate the amount of fuel using the temperature, pressure and previously known volume. Through this, the processor is configured for controlling the opening/closing valve **520** to open the opening/closing valve **520** so that the fuel may be supplied to the storage tank **110**, in which relatively little fuel is stored, through the sub pipeline **510**, when the amount of the fuel accommodated in at least one storage tank **110** is less than the amount of the fuel accommodated in another storage tank **110**.

[0114] However, the present disclosure is not limited thereto, and the amount of the fuel accommodated in the interior of the storage tank **110** may be measured by a separate sensor.

[0115] FIG. **5** is a schematic view of a fuel charging module according to another exemplary embodiment of the present disclosure.

[0116] In the fuel charging module **100-2**, compared to the fuel charging module **100-1** illustrated in FIG. **4**, disposition of the recovery pipeline **500**, the sub pipeline **510**, the opening/closing valve **520**, the sub tank **600**, and the heat-exchange portion **700** may be different.

[0117] According to another exemplary embodiment of the present disclosure, the fuel charging module **100-2** does not recover the fuel from the heat-exchange pipelines **310** and **320** to cause it flow to the charging station **10** (see FIG. **1**) as illustrated in FIG. **4**, and rather, may recover the fuel

from the first and second heat-exchange pipelines **310** and **320** and supply it to the plurality of storage tanks **110**.

[0118] The heat-exchange portion **700** of the fuel charging module **100-2** may be disposed on the recovery pipeline **500** rather than the sub pipeline **510**. That is, according to another exemplary embodiment of the present disclosure, the sub tank **600** may accommodate the fuel that has exchanged heat with the fuel supplied through the receptacle **120**.

[0119] The recovery pipeline **500** may extend from the first and second heat-exchange pipelines **310** and **320** to the sub tank **600**. The sub tank **600** may accommodate the fuel which is recovered through the sub inlet **601** connected to the recovery pipeline **500**. The sub tank **600** may be spaced apart from the heat-exchange pad **200**.

[0120] The heat-exchange portion **700** may be configured to exchange heat between the fuel that flows through the recovery pipeline **500** and the fuel supplied from the charging station **10**. By the heat-exchange portion **700**, the temperature of the fuel that flows through the recovery pipeline **500** may be lowered, and the temperature of the fuel supplied from the charging station **10** may be increased.

[0121] The fuel recovered through the recovery pipeline **500** may be accommodated in the interior of the sub tank **600** after heat is exchanged by the heat-exchange portion **700**.

[0122] The sub tank **600** may be connected to the sub pipeline **510**. The fuel accommodated in the interior of the sub tank **600** may flow through the sub pipeline **510**. The sub pipeline **510** may connect the sub tank **600** and the plurality of storage tanks **110** to supply the fuel contained in the sub tank **600** to the storage tanks **110**.

[0123] The sub pipeline **510** may extend from the sub tank **600** to the plurality of storage tanks **110**, and an opening/closing valve **520** may be provided on the sub pipeline **510**.

[0124] The opening/closing valve **520** may be provided with a solenoid valve, and the like. The opening/closing valve **520** may be opened or closed by a processor so that the fuel accommodated in the sub tank **600** may be supplied to the storage tank **110**, in which relatively little fuel is stored, through the sub pipeline **510**.

[0125] The fuel charging module **100-2** may store the recovered fuel in the sub tank **600** separately from the plurality of storage tanks **110**, and may cause the fuel to flow from the sub tank **600** when the fuel in at least one of the plurality of storage tanks **110** is insufficient or the internal pressure of at least one storage tank is lower than the internal pressure of the other storage tanks.

[0126] By the present principle, a larger amount of fuel may be accommodated than in a structure with only the storage tank **110** in an interior of the vehicle, and a travel distance of the vehicle may be increased by preventing a pressure imbalance between the storage tanks **110**.

[0127] The processor is configured to determine the amount of fuel accommodated in each of the storage tanks **110**, through the temperature of each of the storage tanks **110**. Through this, the processor is configured for controlling the opening/closing valve **520** to be opened so that fuel may be supplied to the storage tank **110**, in which relatively little fuel is stored through the sub pipeline **510** when the amount of the fuel accommodated in at least one storage tank **110** is less than the amount of fuel accommodated in another storage tank **110**.

[0128] However, the present disclosure is not limited thereto, and the flow rate of the fuel accommodated in the interior of the storage tank **110** may be measured by a separate sensor.

[0129] Hereinafter, a fuel charging method will be described with reference to FIG. **6** and FIG. **7**.

[0130] FIG. **6** is a flowchart according to the fuel charging method according to an exemplary embodiment of the present disclosure.

[0131] Referring to FIGS. **1** to **6**, the fuel charging method may include a connecting operation (**S10**), a first fuel supplying operation (**S20**), a temperature measuring operation (**S30**), a first determination operation (**S40**), a supply maintaining operation (**S50**), a tank cooling operation (**S60**), a second determination operation (**S70**), a second fuel supplying operation (**S80**), a third determination operation (**S90**), a flow rate maintaining operation (**S100**), a fourth determination

operation (S110), and a signal transmitting operation (S120).

[0132] The connecting operation (S10) may be an operation of connecting the receptacle **120** to the charging station **10**. The first fuel supplying operation (S20) may be an operation of supplying a fuel from the charging station **10** to the storage tank **110** after the connecting operation (S10).

[0133] The temperature measuring operation (S30) may be an operation of measuring the temperature of the storage tank **110** after the first fuel supplying operation (S20). In the temperature measuring operation (S30), the temperatures of the plurality of storage tanks **110** may be measured with a specific time difference.

[0134] In the temperature measuring operation (S30), not only the temperatures of the storage tanks **110** but also the pressures of the storage tanks **11** may be measured. Accordingly, when the pressure of one of the plurality of storage tanks **110** is more than a predetermined reference pressure, the signal transmitter **130** may transmit a charging stop signal to the charging station **10**. Accordingly, the predetermined reference pressure may be a pressure value excluding a specific error range from 350 bar*1.5.

[0135] The first determination operation (S40) may be an operation of determining the temperature of the storage tank **110** by comparing it with a predetermined reference temperature after the temperature measuring operation. The first determination operation S40 may be an operation of determining whether the temperature of the storage tank **110** reaches or is more than the predetermined reference temperature. Here, the reference temperature may be a temperature excluded from 85 degrees Celsius by a specific error range.

[0136] The supply maintaining operation (S50) may be an operation of maintaining the flow rate of the fuel supplied to the storage tank **110** so that the fuel supplied through the receptacle **120** continues to be supplied to the storage tank **110** when the temperature of the storage tank **110** is less than the reference temperature (No of S40).

[0137] Even after the supply maintaining operation (S50), the first determination operation (S40) of determining whether the temperature of the storage tank **110** reaches the reference temperature or is more than the reference temperature may be performed again.

[0138] The tank cooling operation (S60) may be an operation of controlling the flow of the fuel so that the fuel supplied through the receptacle **120** is directed to the heat-exchange pad **200** when the temperature of the storage tank **110** reaches or is more than the reference temperature (an example of S40). The tank cooling operation (S60) may include controlling the flow of the fuel so that all the fuel supplied through the receptacle **120** is directed to the heat-exchange pad **200**.

[0139] The second determination operation (S70) may be an operation of determining whether the temperature of the storage tank **110** decreases from the reference temperature after the tank cooling operation (S60).

[0140] The second fuel supplying operation (S80) may be an operation of controlling the flow of the fuel to decrease the flow rate of the fuel which is directed to the heat-exchange pad **200** and increase the flow rate of the fuel supplied to the storage tank **110** when the temperature of the storage tank **110** decreases from the reference temperature in the second determination operation (S70) (an example of S70).

[0141] The third determination operation (S90) may be an operation of determining whether the temperature of the storage tank **110** is maintained within a specific range for a predetermined time period after the second fuel supplying operation (S80). Accordingly, maintaining the temperature of the storage tank **110** within a specific range for a predetermined time period may mean that the temperature of the storage tank **110** is measured at a time difference and a difference between the two temperatures is within a specific error range. Here the time difference may be **10** ms.

[0142] The flow rate maintaining operation (S100) may include an operation of controlling the flow of the fuel to maintain the flow rate of the fuel supplied to the storage tank **110** and the flow rate of the fuel which is directed to the heat-exchange pad **200** when the temperature of the storage tank **110** is maintained for a predetermined time period (an example of S90) in the third

determination operation (S90).

[0143] In the third determination operation (S90), when the temperature of the storage tank **110** is not maintained for a predetermined time period (No of S90), the second determination operation (S70) of determining whether the temperature of the storage tank **110** decreases to less than the reference temperature may be performed again.

[0144] The fourth determination operation (S110) may be an operation of determining whether the temperature of the storage tank **110** increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature for a predetermined time period when the temperature of the storage tank **110** does not decrease (No of S70) in the second determination operation (S70). Here, the specific time period may be about 10 s.

[0145] The signal transmitting operation (S120) may be an operation of transmitting a charging stop signal to the charging station **10** when the temperature of the storage tank **110** increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature for a specific time period. Accordingly, it may be determined that there is a problem in the heat-exchange pad **200** or the above-mentioned sensors.

[0146] FIG. 7 is a flowchart according to a fuel charging method according to another exemplary embodiment of the present disclosure.

[0147] The fuel charging method in FIG. 7 is a fuel charging method including a notifying operation (S130) instead of a signal transmitting operation (S120) compared to FIG. 6. Among the operations of the fuel charging method of FIG. 7, for the connecting operation (S10), the first fuel supplying operation (S20), the temperature measuring operation (S30), the first determination operation (S40), the supply maintaining operation (S50), the tank cooling operation (S60), the second determination operation (S70), the second fuel supplying operation (S80), the third determination operation (S90), the flow rate maintaining operation (S100), and the fourth determination operation (S110), the description of FIG. 6 is used.

[0148] Referring to FIGS. 1 and 7, the fuel charging method may include the notifying operation (S130). The notifying operation (S130) may include an operation of providing a notification to the user when the temperature of the storage tank **110** increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature for a predetermined time period (an example of S110) in the fourth determination operation (S110). The specific time period here may also be about 10 s.

[0149] When charging of the charging module **10** begins, the signal transmitter **130** may transmit the temperature of the storage tank **110**, the pressure of the storage tank **110** to the charging station **10** using infrared rays. The processor may constantly monitor the pressure of the storage tank **110**, the temperature of the storage tank **110**, and supplied flow rate of the storage tank **110**.

[0150] When the fourth determination operation is detected by the processor, the first supply pipeline **160**, the inlet **111** are blocked and a charging stop signal may be sent to the charging station **10** to block the hydrogen supply. After this, the phrase “tank temperature abnormal during charging” may pop up on the vehicle cluster, and a related phrase or code may similarly pop up at the charging station.

[0151] According to the fuel charging module **100**, **100-1**, and **100-2** and the fuel charging method described above, because the fuel charging speed may be optimized while the safety of the storage tank **110** is ensured, the fuel charging time may be shortened, and thus, user convenience may be improved.

[0152] This technology may optimize the charging speed of the fuel while the safety of the storage tank is ensured, and thus, the fuel charging time may be shortened.

[0153] Additionally, according to the present technology, the fuel that flows through the heat-exchange pipeline may be reused, and thus, fuel consumption may be lowered.

[0154] Furthermore, the present technology may allow more fuel to be charged to the storage tank through the sub tank and prevent a pressure imbalance between the plurality of storage tanks, and thus, the travel distance of the vehicle may be increased.

[0155] Furthermore, various effects which may be directly or indirectly identified through the present specification may be provided.

[0156] The above description is a simple exemplary description of the technical spirits of the present disclosure, and an ordinary person in the art, to which the present disclosure pertains, may make various corrections and modifications without departing from the essential characteristics of the present disclosure.

[0157] The storage device that can store data which may be thereafter read by a computer system and store and execute program instructions which may be thereafter read by a computer system. Examples of the computer readable recording medium include Hard Disk Drive (HDD), solid state disk (SSD), silicon disk drive (SDD), read-only memory (ROM), random-access memory (RAM), CD-ROMs, magnetic tapes, floppy discs, optical data storage devices, etc and implementation as carrier waves (e.g., transmission over the Internet). Examples of the program instruction include machine language code such as those generated by a compiler, as well as high-level language code which may be executed by a computer using an interpreter or the like.

[0158] In various exemplary embodiments of the present disclosure, each operation described above may be performed by a control device, and the control device may be configured by a plurality of control devices, or an integrated single control device.

[0159] In various exemplary embodiments of the present disclosure, the memory and the processor may be provided as one chip, or provided as separate chips.

[0160] In various exemplary embodiments of the present disclosure, the scope of the present disclosure includes software or machine-executable commands (e.g., an operating system, an application, firmware, a program, etc.) for enabling operations according to the methods of various embodiments to be executed on an apparatus or a computer, a non-transitory computer-readable medium including such software or commands stored thereon and executable on the apparatus or the computer.

[0161] In various exemplary embodiments of the present disclosure, the control device may be implemented in a form of hardware or software, or may be implemented in a combination of hardware and software.

[0162] Software implementations may include software components (or elements), object-oriented software components, class components, task components, processes, functions, attributes, procedures, subroutines, program code segments, drivers, firmware, microcode, data, database, data structures, tables, arrays, and variables. The software, data, and the like may be stored in memory and executed by a processor. The memory or processor may employ a variety of means well-known to a person including ordinary knowledge in the art.

[0163] Furthermore, the terms such as “unit”, “module”, etc. included in the specification mean units for processing at least one function or operation, which may be implemented by hardware, software, or a combination thereof.

[0164] In the flowchart described with reference to the drawings, the flowchart may be performed by the controller or the processor. The order of operations in the flowchart may be changed, a plurality of operations may be merged, or any operation may be divided, and a specific operation may not be performed. Furthermore, the operations in the flowchart may be performed sequentially, but not necessarily performed sequentially. For example, the order of the operations may be changed, and at least two operations may be performed in parallel.

[0165] Hereinafter, the fact that pieces of hardware are coupled operatively may include the fact that a direct and/or indirect connection between the pieces of hardware is established by wired and/or wirelessly.

[0166] In an exemplary embodiment of the present disclosure, the vehicle may be referred to as

being based on a concept including various means of transportation. In some cases, the vehicle may be interpreted as being based on a concept including not only various means of land transportation, such as cars, motorcycles, trucks, and buses, that drive on roads but also various means of transportation such as airplanes, drones, ships, etc.

[0167] For convenience in explanation and accurate definition in the appended claims, the terms “upper”, “lower”, “inner”, “outer”, “up”, “down”, “upwards”, “downwards”, “front”, “rear”, “back”, “inside”, “outside”, “inwardly”, “outwardly”, “interior”, “exterior”, “internal”, “external”, “forwards”, and “backwards” are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood that the term “connect” or its derivatives refer both to direct and indirect connection.

[0168] The term “and/or” may include a combination of a plurality of related listed items or any of a plurality of related listed items. For example, “A and/or B” includes all three cases such as “A”, “B”, and “A and B”.

[0169] In exemplary embodiments of the present disclosure, “at least one of A and B” may refer to “at least one of A or B” or “at least one of combinations of at least one of A and B”. Furthermore, “one or more of A and B” may refer to “one or more of A or B” or “one or more of combinations of one or more of A and B”.

[0170] In the present specification, unless stated otherwise, a singular expression includes a plural expression unless the context clearly indicates otherwise.

[0171] In the exemplary embodiment of the present disclosure, it should be understood that a term such as “include” or “have” is directed to designate that the features, numbers, steps, operations, elements, parts, or combinations thereof described in the specification are present, and does not preclude the possibility of addition or presence of one or more other features, numbers, steps, operations, elements, parts, or combinations thereof.

[0172] According to an exemplary embodiment of the present disclosure, components may be combined with each other to be implemented as one, or some components may be omitted.

[0173] The foregoing descriptions of specific exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described in order to explain certain principles of the invention and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

Claims

1. A fuel charging module comprising: a storage tank configured to store a fuel; a supply pipeline connected to the storage tank and configured to supply the fuel supplied from a charging station, to the storage tank; a flow rate control valve mounted on the supply pipeline; a heat-exchange pad contacting with the storage tank for exchange of heat with the storage tank; and a heat-exchange pipeline connected to the flow rate control valve and the heat-exchange pad, and configured to supply the fuel to the heat-exchange pad, wherein the flow rate control valve is configured to supply the fuel supplied through the charging station, to the storage tank or the heat-exchange pad.
2. The fuel charging module of claim 1, wherein the heat-exchange pipeline is inserted into an interior of the heat-exchange pad.
3. The fuel charging module of claim 1, wherein the storage tank is in plural, wherein the plurality of storage tanks extend in a first direction, and the plurality of storage tanks are disposed in a second direction intersecting the first direction, and wherein the heat-exchange pad includes:

accommodation recesses disposed on opposite sides of the plurality of storage tanks in a third direction intersecting the first direction and the second direction to accommodate the plurality of storage tanks therein; and support portions formed on sides of the accommodation recesses in the second direction to support sides of the storage tanks in the second direction, and into which the heat-exchange pipeline is inserted.

4. The fuel charging module of claim 3, wherein the plurality of support portions are spaced apart from each other in the second direction and the accommodation recesses are interposed between the plurality of support portions, and wherein the heat-exchange pipeline extends to pass through the plurality of support portions in the first direction.

5. The fuel charging module of claim 2, further including: a temperature detecting sensor configured to detect a temperature of the storage tank; and a processor operatively connected to the flow rate control valve and configured to control the flow rate control valve so that a flow rate of the fuel supplied from the charging station to the storage tank is adjusted in response that the temperature of the storage tank reaches a reference temperature.

6. The fuel charging module of claim 5, wherein the processor is further configured to control the flow rate control valve so that supply of the fuel to the storage tank is stopped and the fuel is supplied to the heat-exchange pipeline in response that the temperature of the storage tank reaches the reference temperature.

7. The fuel charging module of claim 6, wherein the processor is further configured to control the flow rate control valve to increase the flow rate of the fuel supplied to the storage tank, and decrease the flow rate of the fuel supplied to the heat-exchange pipeline, in response that the temperature of the storage tank decreases to less than the reference temperature after the flow rate control valve is operated to supply the fuel to the heat-exchange pipeline.

8. The fuel charging module of claim 6, further including: a signal transmitter configured to transmit a signal to the charging station, wherein the processor is further configured to control the signal transmitter to transmit a charging stop signal to the charging station in response that the temperature of the storage tank is increased or maintained for a predetermined time period after the flow rate control valve is operated to supply the fuel to the heat-exchange pipeline.

9. The fuel charging module of claim 1, wherein the heat-exchange pipeline includes one end portion connected to the flow rate control valve, and an opposite end portion extending from the one end portion and formed on an opposite side to the one end portion, and wherein the fuel charging module further includes a recovery pipeline connecting the opposite end portion of the heat-exchange pipeline and the charging station.

10. The fuel charging module of claim 1, wherein the heat-exchange pipeline includes one end portion connected to the flow rate control valve and an opposite end portion extending from the one end portion and formed on an opposite side to the one end portion, and wherein the fuel charging module further includes: a recovery pipeline connected to the opposite end portion of the heat-exchange pipeline; and a heat-exchange portion configured to exchange heat between the fuel recovered through the recovery pipeline and the fuel supplied from the charging station.

11. The fuel charging module of claim 10, further including: a sub tank connected to the recovery pipeline, and spaced apart from the heat-exchange pad to accommodate the fuel recovered through the recovery pipeline; a sub pipeline connecting the sub tank and the storage tank to supply the fuel accommodated in the sub tank to the storage tank; and an opening/closing valve disposed on the sub pipeline.

12. The fuel charging module of claim 10, wherein the heat-exchange portion is positioned on a location, where the supply pipeline crosses the sub pipeline.

13. A fuel charging method comprising: a connecting operation of connecting a receptacle to a charging station; a first fuel supplying operation of supplying a fuel from the charging station to a storage tank; a temperature measuring operation of measuring a temperature of the storage tank; and a tank cooling operation of controlling, by a processor, flow of the fuel supplied through the

receptacle so that the fuel is directed to a heat-exchange pad in response that the temperature of the storage tank reaches a reference temperature or is more than the reference temperature.

14. The fuel charging method of claim 13, wherein the tank cooling operation includes: controlling, by the processor, the flow of the fuel so that all the fuel supplied through the receptacle is directed to the heat-exchange pad.

15. The fuel charging method of claim 13, further including: a second fuel supplying operation of controlling, by the processor, the flow of the fuel supplied to the storage tank to increase a flow rate of the fuel in response that the temperature of the storage tank decreases to less than the reference temperature.

16. The fuel charging method of claim 15, wherein the second fuel supplying operation includes: increasing, by the processor, the flow rate of the fuel supplied to the storage tank, and, by the processor, decreasing a flow rate of the fuel directed to the heat-exchange pad.

17. The fuel charging method of claim 15, further including: a flow rate maintaining operation of controlling, by the processor, the flow of the fuel to maintain the flow rate of the fuel supplied to the storage tank and the flow rate of the fuel directed to the heat-exchange pad in response that the temperature of the storage tank is maintained for a predetermined time period after the second fuel supplying operation.

18. The fuel charging method of claim 13, further including: a signal transmitting operation of transmitting, by the processor, a charging stop signal to the charging station in response that the temperature of the storage tank increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature.

19. The fuel charging method of claim 13, further including: a notifying operation of providing, by the processor, a notification to a user in response that the temperature of the storage tank increases to the reference temperature or higher than the reference temperature or is maintained at the reference temperature or higher than the reference temperature.
